IoT Security in the Quantum Era: State of the Art and Open Challenges

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Moussa DIOP
Departements of computer sciences,
Polytechnic Institute (ESP)
Cheikh Anta Diop University (UCAD)
Dakar, Senegal
moussa71.diop@ucad.edu.sn

Papis NDIAYE
Departements of computer sciences,
Assane Seck University of Ziguinchor (UASZ)
Ziguinchor, Senegal
i.ndiaye@univ-zig.sn

Doudou DIONE
Departements of computer sciences,
Polytechnic Institute (ESP)
Cheikh Anta Diop University (UCAD)
Dakar, senegal
doudou2.dione@ucad.edu.sn

Idy DIOP
Departements of computer sciences,
Polytechnic Institute (ESP)
Cheikh Anta Diop University (UCAD)
Dakar, Senegal
idy.diop@esp.sn

***Abstract*—The Internet of Things (IoT) has grown rapidly, transforming various aspects of daily life into intelligent systems capable of collecting and exchanging data in real time. At the same time, advances in quantum computing have given rise to powerful quantum computers capable of processing complex tasks at unprecedented speed. This disparity between the limited resources of IoT devices and the capabilities of quantum computers poses significant security challenges. Quantum computers introduce new vulnerabilities that threaten IoT systems, particularly in critical applications such as connected healthcare, industrial networks, smart infrastructure, and secure embedded systems. This study reviews the existing literature on post-quantum cryptography (PQC) applied to IoT devices. It analyzes the characteristics and security requirements of IoT environments, current security technologies, and unresolved issues. It highlights the growing threat of post-quantum attacks and the need to adapt security strategies. Finally, the study explores post-quantum code-based cryptosystems as solutions to enhance the security of resource-constrained IoT applications and ensure their resilience against future challenges.**

Keywords*—Internet of Things (IoT), Post-Quantum Cryptography (PQC), Error-correcting code.

I. Introduction

The Internet of Things (IoT) refers to the network of physical objects, such as appliances, instruments, vehicles, buildings, and other elements integrated with electronics, circuits, software, sensors, and network connectivity. This network allows these objects to collect and exchange data. In recent years, IoT has emerged as a promising technology, with many potential applications, ranging from personal IoT to large-scale industrial drone networks. It is estimated that there were 16.7 billion active IoT devices in 2023, a figure that is expected to reach 29 billion by 2027 [1]. However, the security of IoT technologies remains a major issue. Cryptographic algorithms used for encryption and digital authentication can be computationally expensive for resource-constrained IoT nodes, which are often designed to be deployed end masse, sometimes in remote and difficult-to-access areas. Factors such as production cost, size, and battery life are frequently prioritized over security. However, adequate encryption and authentication standards must be put in place for IoT technologies to be effectively integrated into domains such as agriculture, industry, healthcare, military, and daily life.

Currently, one of the main challenges for the security of IoT systems lies in the recent advances in quantum computing. This technological advancement could require the use of even more complex cryptographic algorithms, thus aggravating the problems already faced by IoT systems. For example, Shor's algorithm could enable a sufficiently powerful quantum computer to solve integer factorization and discrete logarithm problems, which are currently the basis of public-key cryptography systems [2]. Furthermore, Grover's algorithm could provide a quadratic speedup for unstructured search, thus facilitating the decryption of symmetric keys [3].

As a result, asymmetric cryptography will have to undergo major changes in response to the emergence of quantum computing, while symmetric encryption systems will have to double their key sizes to maintain their security level. These advances threaten the security of current cryptographic schemes. It is in this context that post-quantum cryptography (PQC) emerges, aiming to provide protection against attacks using powerful quantum computers. PQC should also protect lightweight devices, in order to ensure the security of future IoT systems.

Nevertheless, a significant challenge remains: quantum-resistant encryption and signature systems are generally more computationally intensive than current public-key systems [4]. Thus, the transition to PQC could have negative consequences for IoT systems, whose devices are often limited in terms of power consumption, memory capacity, and processing speed.

Researchers have recently become aware of these difficulties and have undertaken studies to evaluate the performance of PQC algorithms on resource-constrained IoT devices. These evaluations play a crucial role, as they allow IoT manufacturers, system engineers, and security experts to make informed choices regarding the selection and implementation of PQC algorithms. These algorithms must be able to address the risks posed by quantum computing, while meeting the constraints inherent in the IoT environment.

Four major families of cryptographic techniques are currently emerging in the PQC field as potential candidates to protect IoT against quantum risks. These schemes include hash-based, multivariate polynomial-based, lattice-based, and error-correcting code-based schemes. Each of these families presents interesting characteristics and varying feasibility for resource-constrained IoT systems.

The contributions of this study are as follows:

- An in-depth analysis of post-quantum cryptography methods in the IoT domain, taking into account the specific needs of resource-constrained devices.
- A detailed explanation of the characteristics and architectures of IoT devices and their relationship to security, highlighting unsolved challenges related to IoT security.
- A demonstration of the use of post-quantum cryptography for IoT systems and an assessment of potential post-quantum attacks and their impact on the security of these systems.
- An overview of the challenges associated with PQC schemes based on error-correcting codes and future directions for research in this area.

A. *Litetarure review*

Research on IoT security in a post-quantum context has proliferated, exploring both current security challenges and PQC solutions. These studies highlight the opportunities and limitations of existing methods, but also reveal gaps in protecting low-resource IoT devices.

Malina et al. [4] conducted a survey on privacy-preserving technologies in IoT systems, but their information on the PQC candidates selected in the third phase of the NIST competition is outdated. NIST has since selected CRYSTALS-Kyber and CRYSTALS-Dilithium as the best-performing encryption and signature schemes [5]. This study mainly focuses on Kyber1024 and Dilithium-III with data prior to 2019, which limits the relevance of the results to recent advances. Asif et al. [6], in a study conducted in an academic environment, gathered extensive data on attack vectors and the importance of IoT in organizations. However, their approach relies on classical methods and basic hashing techniques, highlighting the need to explore more advanced and quantum-specific schemes. Guillen et al. [7] analyzed the effectiveness of NTRUEncrypt for battery-powered IoT devices, especially on ARM Cortex-M0 microcontrollers, and demonstrated advantages in terms of memory and energy consumption. However, the NTRUEncrypt scheme has security tradeoffs compared to more recent alternatives, which requires rigorous evaluations for critical applications. Ebrahimi et al. [8] introduced InvRBLWE, an improved version of Ring-LWE, demonstrating good performance for FPGAs and ASICs in terms of speed and efficiency. This solution is distinguished by its stability against quantum attacks, but the scalability of these results to resource-constrained IoT devices remains to be validated. Furthermore, Mayes et al. [9] implemented Kyber768 on a Trust-Anchor MULTOS chip for IoT, using a Kronecker multiplier to improve performance, showing increased efficiency in IoT environments, although the application is limited to specific configurations. Saribas et al. [10] compared three key wrapping mechanisms and two digital signature algorithms, showing that post-quantum methods introduce overhead but remain promising for TLS 1.3-like protocols. However, these solutions often lead to high power consumption, which is problematic for low-power IoT devices. Similarly, Liu et al. [11] highlighted the performance differences between ATmega12841 and ARM Cortex-M4F microcontrollers, with the former offering higher encryption speed but lower security. However, the study does not examine key authentication protocols for low-resource IoT environments in detail. Fernandez-Carame [12] and Khalid et al. [13] have highlighted the challenges and future trends of IoT security in the context of increasing quantum threats, while highlighting the specific security risks of current IoT standards and the importance of quantum-resistant cryptosystems. Finally, other researchers, such as Schöffer et al. [14] and Kumar and Garhwal [15], have proposed hybrid solutions and lattice-based cryptosystems, indicating that these methods could offer increased post-quantum security, but with challenges related to computational complexity and resource management.

Current PQC approaches for IoT include various lattice-based cryptographic schemes (e.g., Ring-LWE, Kyber) and code-based cryptosystems (e.g., code-based cryptosystems) that promise resistance to quantum attacks. While these methods offer advantages, such as cryptographic robustness and long-term protection, they often introduce computational overhead and memory requirements, making their integration difficult for resource-constrained IoT devices (low power, limited memory). Some schemes, such as NTRUEncrypt and hybrid methods, have proven to be more suitable for commonly used IoT microcontrollers, but may still pose scalability and power consumption challenges for distributed IoT environments.

The proposed method distinguishes itself by its code-based cryptosystem approach, specifically tailored for low-resource IoT devices, providing proven quantum resistance while reducing power consumption and computational complexity. Unlike other studies that focus on generalized post-quantum solutions, our approach targets microcontrollers and embedded systems, maximizing efficiency for critical applications where energy conservation and lightweight are essential. By adopting this method, we propose a cryptographic framework that is tailored to IoT constraints while ensuring future-proof security against quantum computing threats. This approach opens up prospects for protecting critical infrastructure and sensitive data in a future where quantum-computing capabilities could compromise classical cryptographic schemes.

II. IoT Security

This section describes the essential features of the IoT architecture, security issues according to the layers, and open problems related to security [16] - [21]. In IoT, objects facilitate communication, whether interactive or not, with the following main characteristics: dynamic and varied structure, various forms of communication, high computing capacity, data management, low need for portability, remote control, environmental monitoring, use of different layered technologies, and dynamic network infrastructure with sensors and actuators.

The IoT architecture must be explained to facilitate understanding and define operations performed in the IoT environment formed by the combination of many objects.

A. IoT Architecture

In this subsection, we recall three (3) main layer of the IoT and security requirements in the IoT architecture. The main components and protocols of the IoT layer are illustrated in Fig.1.

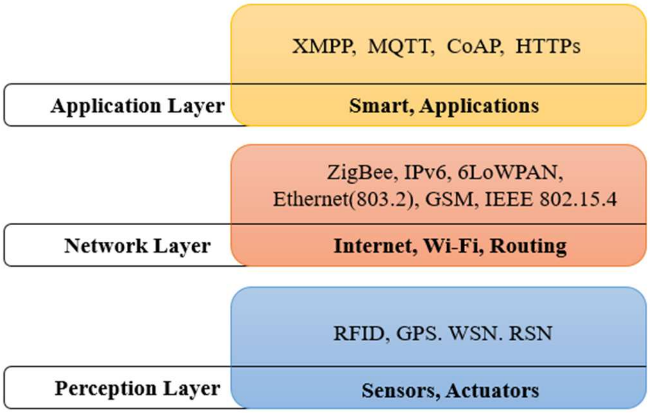

Fig. 1. IoT Architecture

The essential layers and features of IoT are summarized below:

a) Perception layer

The perception layer is the first step of the IoT architecture [18], [22]. It relies on sensors and actuators to collect and interact with environmental data, such as temperature, humidity or brightness. Technologies such as radio frequency identification (RFID), global positioning system (GPS), wireless sensor networks (WSN) or RFID sensor networks (RSN) are widely used in this layer. However, sensor nodes, limited in computing power and storage, struggle to integrate robust security mechanisms, making this layer vulnerable to attacks. Adapted solutions, such as lightweight cryptographic algorithms, are necessary to protect data. Furthermore, the integration of anti-malware systems, multi-factor authentication mechanisms, key management and anomaly detection is essential to strengthen the security of this critical layer.

b) Network layer

The communication layer [18], [22], is essential in the IoT architecture, as it ensures the transmission and storage of data coming from the perception layer. It uses network topologies such as star, which centralizes communication but is vulnerable to single failures, and mesh, which provides redundancy and resilience. Common protocols include IPv6, Wi-Fi, GSM, 6LoWPAN, Ethernet, IEEE 802.15.4, and ZigBee, each tailored to specific needs. However, this layer faces challenges such as scalability, protocol diversity, and device mobility, making it vulnerable to attacks such as denial of service or data interception. To enhance its security, advanced machine learning techniques, such as deep neural networks and anomaly detection algorithms, are used to identify threats and optimize network management. These approaches provide adaptive protection against evolving threats

c) Application layer

IoT applications span diverse domains, such as smart grids, cities, factories, transportation, and healthcare. They enable data generation, management, and processing, as well as notification, alerting, and control functions. The middle layer, integrated with the application layer, facilitates data sharing through technologies such as machine-to-machine (M2M), cloud, and service platforms. Security challenges include privacy, sharing, and access control. Protocols such as Constrained Application Protocols (CoAP), Message Queue Telemetry Transport (MQTT), Extensible Messaging and Presence Protocol (XMPP), and Hypertext Transfer Protocol Secure (HTTPS) are used to address these challenges. However, these solutions must adapt to the constraints of IoT devices, such as hardware and software limitations. Autonomous and self-protecting approaches integrating control and analysis mechanisms enhance security and fill gaps in this layer [19], [23].

B. IoT Communication Protocols

IoT data protocols, illustrated in Fig. 1, play a crucial role in facilitating communication and interaction between various IoT devices. They enable smooth and efficient communication across the entire IoT ecosystem, which includes sensors, actuators, appliances, gateways, user applications, and servers. Through these protocols, the interactive and integrated communication model between these different components becomes possible, allowing for optimal management and utilization of data and resources. These protocols and their usage are listed below [16], [24], [25]:

- MQTT (Message Queuing Telemetry Transport): A lightweight TCP/IP-based protocol designed for environments requiring low power consumption and reduced bandwidth. Ideal for IoT sensors and battery-powered devices due to its efficiency in transferring small messages.
- CoAP (Constrained Application Protocol): A simplified version of the HTTP model that runs on UDP (User Datagram Protocol), suitable for IoT devices with limited resources. This protocol enables efficient and secure communication in constrained environments.
- ZigBee: A wireless protocol that runs on the 2.4 GHz band, suitable for applications requiring low power consumption and enhanced security. Its efficient management of mesh networks makes it ideal for home and industrial sensor networks.
- 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks): Integrates IPv6 into low-power devices using compression techniques to reduce header sizes. Crucial for connecting IoT devices in constrained wireless networks.
- RFID (Radio-Frequency Identification): A wireless technology that identifies and tracks objects using RFID tags and readers. Commonly used for inventory management, asset tracking, and process automation in the IoT.

- WSN (Wireless Sensor Networks): Autonomous sensor networks that offer flexible and scalable data collection. Used for environmental monitoring, infrastructure management and security systems, with rapid and adaptable installation.

These protocols ensure communication between different IoT components, each with its own specific advantages and applications. Their use not only allows efficient data management, but also a harmonious integration of devices in various environments and applications, thus contributing to the overall efficiency and performance of IoT systems.

C) *IoT Devise Attacks*

IoT applications, due to their interconnectedness and often distributed architecture, are vulnerable to several types of attacks [17], [18], [20], [21], [25]:

- Denial of Service (DoS) and DDoS attacks: Overload IoT networks to make services inaccessible.
- Man-in-the-middle (MITM) attacks: Intercept communications between IoT devices to steal or modify sensitive data.
- Packet injection attacks: Inject malicious packets into the network, disrupting normal operations.
- Device Hijacking: Take control of IoT devices, often to use them in broader attacks such as botnets.
- Software Exploitation: Target IoT devices with insecure or outdated software or firmware to execute malicious code.
- Attacks on sensors and actuators: Aim to falsify or alter the data collected, affecting the accuracy of dependent systems.
- Data theft: Access sensitive data transmitted or stored, such as personal information or critical data.
- Privacy violation: Collect information about users without their consent, exposing their privacy.
- Eavesdropping attacks: Intercept unencrypted communications to obtain sensitive information.
- Compromise of IoT management systems: Target the central platforms that control the devices, compromising the integrity of the network.
- API attacks: Exploit the management APIs of IoT devices if they are not properly secured.
- Cross-site scripting (XSS) and SQL injection attacks: Target the web interfaces used to manage IoT devices, compromising their security.
- Unauthorized physical access: Physically manipulate publicly accessible IoT devices to compromise their security.
- Side-channel attacks (SCA): Use unintentionally leaked information to extract cryptographic keys or other sensitive information.
- Brute force and dictionary attacks: Target IoT devices with weak cryptographic security mechanisms or keys that are too short.
- Reverse engineering attacks: Analyze IoT devices to discover and exploit their cryptographic algorithms.
- Post-quantum attacks: With the advent of quantum computing, some current cryptographic systems may become vulnerable; these attacks will be detailed in Section 4.5.

D) *IoT Appication Security Technique*

To ensure the security of IoT environments and applications, several methods are used [19], [26] - [28]:

- **Blockchain:** Facilitates the connection of IoT applications to the cloud while ensuring data security and confidentiality thanks to its distributed structure. Blockchain [29], [30] uses hash values, verifies data with miners to avoid corrupted data, and protects communications with private and public keys. For IoT devices with limited resources, a proxy-based architecture is employed to manage blockchain ledgers. This technology also helps prevent DDoS attacks and single-point outages.
- **Fog computing:** Introduced by Cisco in 2012, supports cloud computing by reducing data storage in the cloud and improving security. For IoT, it aims to integrate data-generating objects and increase the volume, variety, and velocity of data. Unlike cloud computing, fog computing places resources closer to end users, thereby reducing latency. It helps prevent various types of attacks and data transfer issues in IoT environments [19], [34].
- **Machine Learning (ML):** enhances the security of IoT networks by analyzing device behaviors and quickly detecting attacks. Although challenges remain, such as managing energy, memory, and computational requirements [23], [32], [33], ML is widely used for authentication, intrusion prevention, and detection of attacks, fraud, and malware.
  IoT devices generate vast volumes of data, which ML models leverage to detect anomalies, predict malicious behavior, and improve system resilience.
  - Supervised intrusion detection: A recent study [34] shows that supervised models such as support vector machines (SVMs) and random forests effectively detect IoT-specific attacks, such as jamming or DDoS attacks, with high accuracy.
  - Optimized ensemble models: Another approach, based on the Gray Wolf Optimizer, improves intrusion detection by combining multiple classifiers [35]. This method reduces false positives and improves performance in complex environments.
  - Hybrid approaches for industrial systems: A research [36] proposes a hybrid method using an autoencoder, deep convolutional networks (DCNN), and long short-term memory (LSTM) networks. This approach proactively detects anomalies in critical industrial control systems, by efficiently processing sequential data.

  By integrating these techniques, ML provides innovative solutions to counter complex threats and improve IoT security, while offering promising prospects for the future.
- **Edge Computing:** Uses powerful computing devices close to IoT objects to reduce the computational costs related to advanced security [19], [29], [37]. This approach is distinguished by its ability to process data locally, close to the sources, eliminating transmission

delays to the cloud. By processing data at the edge, sensitive information is less exposed to attacks while in transit, reducing the risk of data breaches. In addition, proximity allows for rapid detection and response to threats, while minimizing latency in critical systems. The edge also supports sophisticated encryption mechanisms and ensures secure connections while meeting the power and processing limitations of IoT devices. This technique optimizes real-time data management and strengthens the security of IoT applications by providing a decentralized architecture that reduces single points of failure.

In general, security issues such as data breaches and compliance issues that can occur in IoT applications are prioritized to be addressed through these innovative approaches.

E) IoT Security Challenges

To ensure the security of IoT functions and communications, several key issues need to be addressed [12], [17], [19], [23], [29] - [39]:

- Preventing potential attacks due to the diversity of IoT devices with varying security requirements.
- Assessing identification needs to improve the effectiveness of machine learning methods in IoT security.
- Investigating data privacy and security challenges in remote computing methods.
- Identifying challenges related to blockchain usage, including storage costs, distribution speed, scalability, and usability.
- Analyzing security issues of data storage, device registration, encryption, and authentication.
- Developing technologies to secure gateways in layered IoT architecture.
- Creating encryption protocols for end-to-end secure communication despite resource limitations.
- Integrating trust management via digital signatures and authentication into IoT applications.
- Adapting security mechanisms to match the diversity of IoT devices.
- Realizing efficient cryptographic systems to ensure security while optimizing energy consumption.
- Optimizing key generation in public-key cryptography systems to minimize resource usage.
- Considering security constraints in IETF-defined restricted node specifications.
- Building quantum-resistant cryptography systems, in anticipation of future threats posed by quantum computers.
- Analyzing the feasibility of quantum-resistant cryptosystems in IoT devices.

These efforts aim to improve security while taking into account the constraints of IoT devices.

III. IoT Security in the Quantum Era

A. Quantum Computing

Quantum computing is based on the properties of qubits, particles that can exist in a superposition of multiple states simultaneously. Unlike classical bits (0 or 1) [40], qubits allow exponentially faster calculations thanks to phenomena such as superposition and entanglement. These properties allow quantum computers to solve complex problems that are impossible or very long to process for traditional computers.

B. Quantum Threats to IoT Security

Much work in quantum computing aims to develop efficient algorithms capable of solving complex tasks, such as search problems. These algorithms, although powerful, pose serious threats to the security of IoT systems.

- Shor's Algorithm [2]: This algorithm allows numbers to be factored into their prime components exponentially, thus compromising asymmetric cryptography systems that rely on the difficulty of this operation.
- Grover algorithm [3]: It optimizes unstructured searches by reducing their time complexity to $O(\sqrt{N})$, compared to $O(N)$ for classical algorithms, which significantly improves the efficiency of searches on unsorted databases.
- Simon algorithm [41]: This algorithm identifies whether a function is one-to-one $(1:1)$ or one-to-one$(2:1)$, with important applications in cryptographic analysis.

In IoT systems, symmetric and asymmetric cryptographic algorithms protect data but remain vulnerable to quantum attacks, notably via Shor's algorithm, which can break current cryptographic schemes. To counter this threat, quantum computing offers solutions such as quantum key distribution (QKD) and quantum key recycling (QKR). QKD uses quantum properties to distribute secure keys, while QKR allows encryption keys to be reused without generating new ones, thus reducing resource requirements. An 8-state coding approach facilitates its implementation without a quantum computer. The transition to quantum cryptographic algorithms is essential to ensure the security and privacy of IoT devices against future quantum threats.

C. Post-Quantum Cryptography (PQC)

Post-quantum cryptography (PQC) aims to ensure the security of data and communications against threats from quantum computers. These algorithms are based on complex mathematical problems that are supposedly difficult to solve by these machines [42]. Their development, analysis and standardization are underway to ensure a gradual transition to a secure infrastructure resistant to quantum attacks [43]. A key component of PQC is post-quantum key distribution, which protects against attacks on conventional asymmetric cryptography systems [44]. Integrated into current protocols, this technology offers a future solution to secure key exchanges, even with advanced quantum computers.

IoT devices, often resource-constrained, are particularly vulnerable to these threats. To protect them, four cryptographic approaches are emerging as potential solutions for securing IoT Fig. 2: hashing, code, multivariate, and network.

- **Hash-based cryptography:** Uses algorithms like XMSS, which rely on collision-resistant hash functions. They are efficient for constrained devices [55].

- **Multivariate cryptography:** Schemes like UOV, based on systems of multivariate polynomial equations, provide IoT-friendly security with small keys and low resource consumption [46].
- **Network-based cryptography:** Schemes like NTRUEncrypt or NTRUSign, known for their resistance to quantum attacks, are flexible and efficient, with IoT-friendly key sizes [47].
- **Code-based cryptography:** Algorithms like McEliece use the complexity of error-correcting codes to ensure security. Despite their large keys, optimizations are underway to make them IoT-friendly [48].

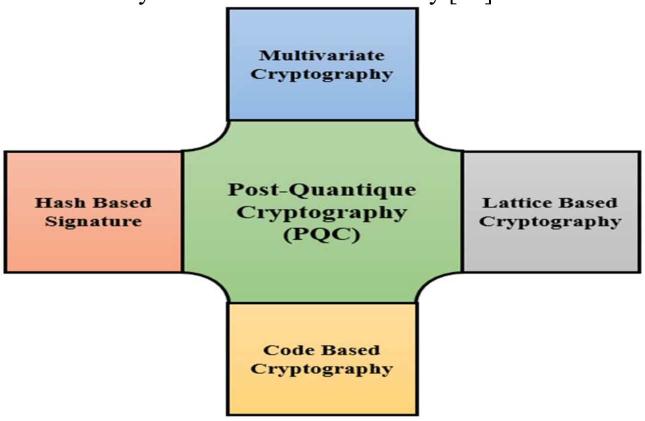

Fig. 2. Type of post-quantum cryptography

These solutions, adapted to the constraints of IoT, make it possible to ensure a transition towards reinforced security in the quantum era.

D. Vulnerabilities of IoT in the Quantum Era

The vulnerabilities of IoT systems in a post-quantum context are discussed in two points. First, the development of secure and efficient post-quantum cryptographic solutions will require years of research. Therefore, it is essential to quickly find optimal solutions. Second, the transition from classical cryptographic infrastructure to a quantum infrastructure will take time [15]. Therefore, it is more efficient to research security solutions with current resources while preparing the migration to quantum cryptography [49]. The IoT is particularly vulnerable to threats, especially those related to quantum computing. The current infrastructure has weaknesses in confidentiality, integrity, and availability, especially due to weak encryption. Cryptographic algorithms used in IoT systems suffer from two major problems: the small size of encryption keys and the difficulty in handling large factorizations and discrete logarithmic equations. This exposes IoT systems to Shor factorization attacks, which can compromise sensitive data [50]. Another important vulnerability of IoT systems is the use of weak or hard-coded passwords, often publicly accessible or introduced through backdoors in software. This allows attackers to easily access IoT devices, create botnets, or spread malware [51]. Password management in a distributed IoT environment is complex and time-consuming, making it an easy target for attackers. Moreover, unnecessary network services on IoT devices, especially those connected to the Internet expose systems to risks such as MITM (man-in-the-middle) attacks, aimed at stealing sensitive information or controlling devices remotely. Insecure ecosystem interfaces, such as websites, backend APIs, and cloud or mobile interfaces, facilitate IoT device compromises due to lack of authentication, weak encryption, and inadequate input and output management [52]. Additionally, the lack of secure update mechanisms, such as insufficient firmware validation, unencrypted transmission, and lack of anti-rollback procedures, expose devices to attacks [52], [53]. Pirated updates can lead to the failure of critical devices in industries such as healthcare and energy. The use of insecure or outdated components, including vulnerable software or libraries, exposes IoT devices to hacks. This includes dangerous customizations of operating systems and the use of components from corrupt supply chains [49]. Outdated software, especially open source components, may contain known vulnerabilities to attackers, making IoT devices more vulnerable [53]. Additionally, insecure data transfer and storage, with weak encryption and lack of access control, facilitate attacks on sensitive data in transit or in process. Lack of IoT device management, including asset management, updates, secure deactivation and system control, represents a major security weakness [49]. Poor device management throughout their lifecycle allows unauthorized devices to access networks and sensitive information [52]. Additionally, the lack of physical hardening exposes IoT devices, used in remote locations, to direct attacks. These vulnerabilities, such as weak passwords, insecure data, unprotected interfaces, and lack of secure updates, are exploited by quantum computing to launch attacks on IoT systems. Finally, attackers can leverage these weaknesses to conduct quantum key recovery attacks and further compromise classical systems [54].

This section highlights the main weaknesses of IoT systems in a quantum environment. First, it highlights that developing secure cryptographic solutions suitable for quantum computing requires a significant investment in time and resources. Moreover, the transition to a quantum cryptographic infrastructure is a lengthy process. The emphasis is on the urgency of finding effective security solutions using currently available resources, to protect IoT systems throughout this transition.

E. Quantum Attacks on IoT

In this section, we will study possible post-quantum attacks on IoT systems in terms of IoT layers.

a) Perception Layer

Quantum attacks on the perception layer include jamming attacks, DDoS attacks, and quantum desynchronization attacks. Among these attacks are quantum fake state attacks and quantum Trojan attacks. In [55], the authors describe the quantum fake state attack, where the attacker (Eve) generates fake states or light pulses that are embedded in the communication between Alice and Bob. What makes this attack particularly insidious is that, despite the introduction of these deceptive states, the error rate that is a potential indicator of compromise remains unchanged. This subtle manipulation allows Eve to mask the attack, making it difficult for Alice and Bob to detect anomalies during the quantum key distribution, thus increasing the sophistication and evasiveness of the adversary. In [56], the authors propose a quantum Trojan horse attack scenario, where Eve attacks the communication between Alice and Bob by injecting a high-intensity light pulse containing Trojan horse photons into the optical fiber. When Alice encodes her photons for transmission to Bob, she unintentionally encodes Eve's photons, as she is unaware of the attack in progress. The Trojan horse photons, reflected by the elements on Alice's side, travel to Bob. Eve intercepts these photons upon their arrival on Alice's side, allowing her to determine the information encoded by Alice during this covert attack. Jamming attacks typically occur in the perception layer, especially in WSNs. These attacks render the communication medium inoperable by sending many requests or disrupting the communication to block responses [57]. In the quantum version of this attack, the entanglement property between the sent qubits is exploited to modify the communication and eavesdrop on the traffic. Entanglement allows each pair of entangled qubits to correlate their states, which accelerates the capture of the encryption key and allows attackers to send excessive requests to damage the communication channel. In [50], a survey on the main threats and challenges of IoT in a post-quantum world proposes a quantum desynchronization attack against the perception layers of IoT devices. This attack uses malicious entangled qubits to block communication between the parties, thereby damaging the WSNs and preventing them from sending or receiving data. The authors of [58] developed a quantum computing-inspired security ensemble model to detect quantum DDoS attacks at the perception layers of IoT devices. Unlike the classical DDoS attack, the quantum version sends malicious qubits that may be in a superposition state, which results in sending large amounts of malicious data that the perceptions cannot process, thus blocking the secure connection. The proposed security solution against this attack uses a quantum protocol to manage secure communication at the data plane, and a machine learning-based ensemble classifier to detect DDoS attack traffic at the control plane.

b) Network Layer

Quantum network attacks on IoT devices include various forms such as quantum insertion attacks, quantum key recovery attacks, quantum man-in-the-middle attacks, quantum saturation attacks, local oscillator (LO) intensity attack, and quantum calibration attack. Quantum insertion attack, also known as HTML redirection attack, targets communication protocols such as TCP. In this attack, the attacker injects malicious content into a specific TCP session using selectors such as tracking cookies to maintain control over the network [59]. Fig. 3 illustrates how this attack can take control of a TCP communication session. The attacker needs to obtain the source and destination IP addresses, port numbers, and sequence and acknowledgement numbers to inject a fake TCP packet before the server acknowledgement arrives. By using quantum entanglement, the attacker can accelerate the arrival of the malicious packet. To counter this attack, one can analyze the packets in response to a service request to distinguish legitimate packets from malicious packets or use the packet time-to-live (TTL), with infected packets having a shorter TTL [51], [59]. Quantum key recovery attacks combine Shor and Simon quantum algorithms to compromise secure communication channels. Simon algorithm is used to identify the encryption function, while Shor algorithm is employed to extract the secret key, jeopardizing the security of IoT systems [60]. In [61], the authors propose a quantum man-in-the-middle attack targeting secure sessions in cryptocurrency networks. This attack uses quantum entanglement to capture gateway packets and trick victims into believing that the gateway MAC address is that of the attacker, thereby compromising the secure session. This attack can be mitigated by using an end-to-end post-quantum encryption solution, combining the Interplanetary File System (IPFS) and Ethereum contracts to ensure secure session keys. Quantum saturation attack on Continuous-Variable Quantum Key Distribution (CVQKD) is an active side-channel attack that affects the Gaussian modulated coherent state of the CVQKD protocol. This attack combines an intercept-forward with a saturation of the homodyne detection to compromise the secure communication channel. The radical post-selection and Gaussian post-selection solutions are proposed to mitigate this attack. The radical post-selection post selection the measurement results within a confidence interval, but gives the attacker the ability to control the shift, which compromises the security of the channel. The Gaussian post-selection improves on this approach by avoiding the shift control, although it is complex to implement [62]. Finally, in [63], the authors present the quantum calibration attack on CVQKD. This attack uses partial intercept-forward (PIR) to intercept quantum measurements and modify the structure of LO pulses, thereby compromising communication and shared sensitive data.

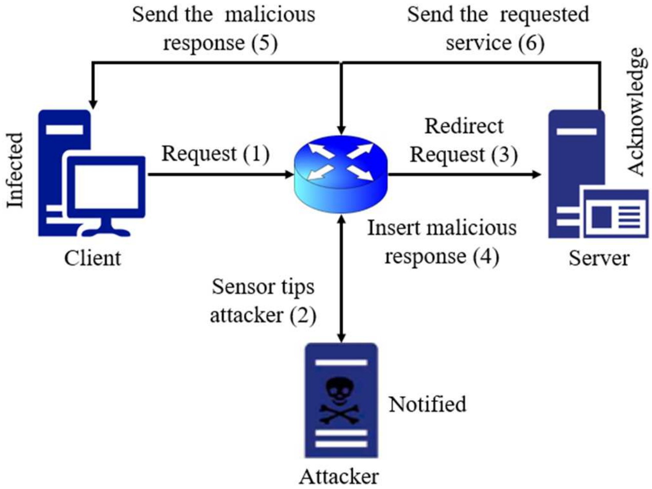

Fig. 3. Quantum insertion attack

c) Application Layer

Quantum attacks on the application layer include attacks on bitcoins, cloud containers, and blockchain, including quantum state attacks. The authors of [64] studied quantum attacks targeting bitcoins, particularly those targeting the Elliptic Curve Digital Signature Algorithm (ECDSA) used to secure transactions. [65] Proposed a quantum laser damage attack, where an attacker manipulates optical pulses directed to the communication channel, damaging the avalanche photodiode of the detector. This attack exploits flaws in the detector, which can harm the ECDSA algorithm while leaving the overall error rate unchanged. This vulnerability can also affect other elements on the legitimate user side. To counter this attack, end-to-end post-quantum encryption is recommended [61]. Another attack method involves Shor's algorithm to capture the private key associated with a public key broadcast by the ECDSA algorithm, allowing a cybercriminal to conduct fraudulent transactions from the compromised address. This threat can also be mitigated by post-quantum encryption [61]. In [66], the authors proposed the quantum hammer attack, an encryption method that combines a bit-tracing attack enabled by Row hammer fault injection with a divide-and-conquer attack. This attack uses the Row hammer bit tracing to extract bits from the secret keys and the divide-and-conquer approach to solve the secret key equation systems more efficiently, thereby compromising the ECDSA algorithm. The quantum key recycling (QKR) technique, which uses 8-state encoding, can be used to generate secure encryption keys and reuse them during login, providing protection against this attack [67]. Quantum attacks on cloud container platforms have been addressed by [68], who also proposed a quantum solution framework to counter them. These threats, such as DDoS attacks, aim to compromise the host among independent containers to gain root access to the host kernel, thereby exposing the entire system. To prevent these attacks, [68] developed a quantum network security protocol that uses a secure quantum channel with a one-time policy, ensuring that key quantum information cannot be regenerated without detection. In their study [69], the authors developed a quantum blockchain scheme to improve blockchain security in the context of IoT. Their approach uses quantum entanglement and delegated proof of stake (DPoS) to secure blockchain keys and transactions. Tests have shown that this solution is effective against quantum state attacks, such as man-in-the-middle attacks, which seek to crack quantum cryptography algorithms by changing the state of secret keys to make them vulnerable. The security analysis indicates that the proposed solution can counter these types of attacks.

After analyzing quantum security attacks on the layers of IoT systems, it is possible to highlight the complex vulnerabilities present in various domains. Examining the threats on the different layers of IoT systems reveals the diversity of risks associated with quantum computing. It is therefore crucial to overcome these major limitations in order to develop effective and robust solutions to protect IoT devices against quantum attacks.

F. *Post-Quantum IoT Security*

This subsection summarizes the changes that resource-constrained IoT devices will undergo in the era of quantum computers based on security requirements. Today, IoT devices use traditional cryptosystems such as RSA (Rivest-Shamir-Adleman), ECC (Elliptic Curve Cryptography), and Diffie-Hellman key exchange to ensure confidentiality and authentication. However, these systems are directly threatened by Shor's algorithm, which enables fast integer factorization and the discrete logarithm problem on a large-scale quantum computer. In contrast, symmetric algorithms such as AES (Advanced Encryption Standard) and hash functions (SHA-2, SHA-3) remain relatively secure, provided that key sizes are increased (e.g., AES-256 rather than AES-128). However, their performance in resource-constrained IoT environments represents a major challenge. In response to these challenges, NIST launched PQC standardization process in 2017 to identify robust solutions against potential quantum attacks. After several evaluation phases, four algorithms were selected for future adoption in 2022:

- CRYSTALS-Kyber (key establishment),
- CRYSTALS-Dilithium, FALCON, and SPHINCS+ (digital signatures).

In addition, schemes based on error-correcting codes, such as Classic McEliece, BIKE, and HQC, remain under evaluation.

Integrating these solutions into IoT environments poses various challenges due to computational, memory, and power consumption constraints. Optimizing key size and resource requirements is essential to ensure effective security without compromising performance. According to Section 2.1 of the literature review, network-based schemes, such as Kyber and Ring-LWE, appear to be the most suitable. However, research is ongoing, particularly on code-based cryptosystems, which represent a promising alternative to ensure IoT security in the post-quantum era.

IV. CODE-BASED CRYPTOSYSTEM FOR IOT

To protect IoT devices from threats posed by quantum computers, various cryptosystems based on error-correcting codes stand out as robust and suitable solutions. These systems, resistant to post-quantum attacks, are essential for IoT devices, which are often limited in resources.

- McEliece cryptosystem: Uses linear codes such as Goppa codes, ensuring high security against quantum attacks. Its main challenge is the size of public keys, requiring optimizations for use in IoT.
- LDPC-based cryptosystems: Offer efficient error correction and more compact keys, making them attractive for resource-constrained IoT devices.
- Polar-based cryptosystems: These recent codes, close to the Shannon limit, show interesting potential for IoT cryptography, although their post-quantum security requires further research.
- Cyclic and quasi-cyclic code-based cryptosystems: Offer a good balance between security and performance with efficient implementations and smaller keys, ideal for IoT.
- Hybrid cryptosystems: Combine error-correcting codes with other post-quantum techniques, providing enhanced security while optimizing efficiency and key size for IoT devices.

These approaches are part of a strategy to secure communications and meet the specific requirements of critical IoT environments.

A. *Challenges of Code-Based Cryptosystems*

Quantum computing and cryptography algorithms pose various security challenges to protect the Internet of Things (IoT). Specific challenges of code-based cryptography include:

- **Key size:** One of the main challenges of code-based systems is the size of public keys, which can be hundreds of kilobytes or more. This can pose storage and transmission challenges for IoT devices with limited resources.
- **Implementation complexity:** Integrating code-based cryptography into existing IoT systems requires software and hardware adjustments. This can be challenging, especially for already deployed devices that cannot be easily updated.
- **Bandwidth requirements:** The size of keys and signatures can also increase bandwidth requirements for secure communications, which can be problematic in environments with limited network connectivity.
- **Post-quantum transition:** Although code-based cryptography is expected to be resilient to attacks from quantum computers, it may still face challenges in terms of transitioning from traditional cryptographic algorithms to code-based schemes

B. *Perspectives*

With the advances in quantum computers, classical cryptographic schemes are likely to become vulnerable to quantum attacks, while the integration of post-quantum cryptography still presents some challenges. Most recent research focuses on adapting quantum cryptography to Internet of Things (IoT) devices, which poses specific challenges due to the resource limitations and reduced computational capabilities of these devices. Future directions for the use of post-quantum code-based cryptography include:

- Focusing on creating efficient implementations of code-based cryptography specifically designed for IoT devices, taking into account their resource constraints.
- Investigating lightweight cryptographic schemes that combine security with computational efficiency would be beneficial for IoT applications.

V. Conclusion and Future Work

The rapid expansion of IoT devices brings increasing security challenges, accentuated by the emergence of quantum threats. Classical cryptosystems, although effective to some extent, need to be complemented by post-quantum cryptosystems (PQC) to protect data and communications in an IoT environment. This study explored the integration of PQC in IoT, highlighting the advantages of code-based systems, such as McEliece, while highlighting the challenges related to their efficiency and compatibility with resource-constrained devices.

To strengthen this transition, several future research areas are identified:

- Optimization of PQC schemes for IoT: Adapt current algorithms to the constraints of IoT devices (energy, memory) by improving schemes such as LDPC and hybrid codes.
- Performance evaluation on various microcontrollers: Study the performance of PQCs on hardware platforms such as ARM Cortex or ATmega to identify configurations suitable for different IoT applications.
- Design of specific security protocols: Develop lightweight versions of common IoT protocols (TLS, DTLS) integrating PQCs to ensure secure connections in distributed environments.
- Reduction of key and signature sizes: Explore methods to compress and reduce the size of PQC keys and signatures, which are crucial for resource-constrained IoT devices.
- Risk assessment in critical applications: Test the resilience of PQCs in critical contexts such as medical or industrial devices, by simulating quantum attacks.
- Development of PQC standards for IoT: Collaborate with standardization bodies such as NIST to establish standards adapted to the specific needs of IoT, ensuring security and efficiency against quantum threats.

This work will contribute to securing IoT infrastructures in a world where quantum computing could disrupt current cryptography paradigms.

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